

❖ **Section 6.** Periodicity, Absorbing, Ergodic, Reversibility

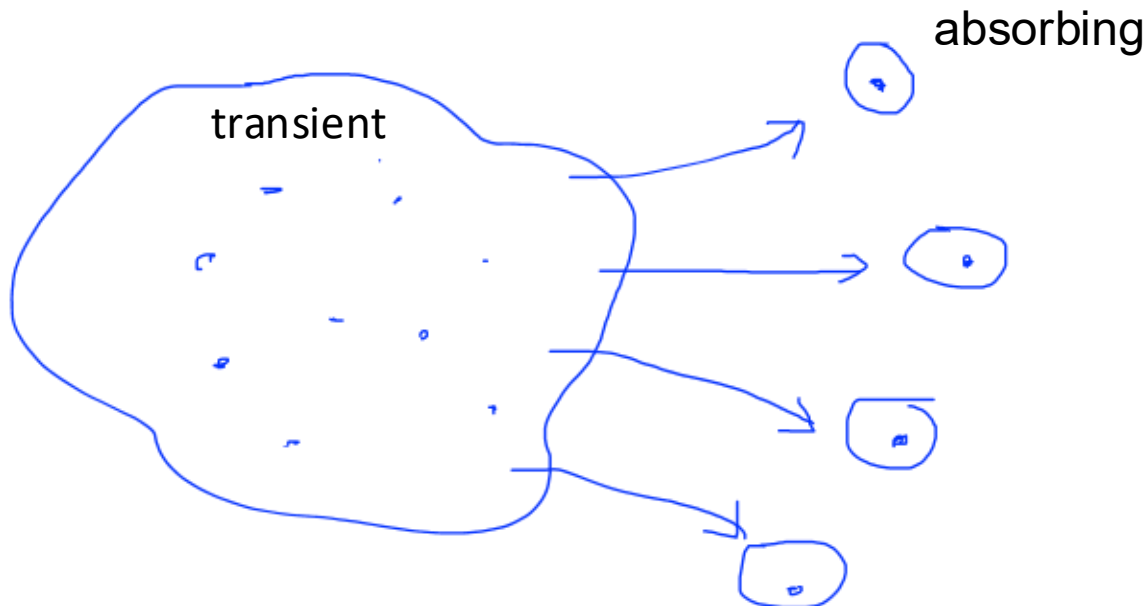
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- Absorbing Chains
- Periodic Chains
- Ergodic Chains
- Time Reversible Chains

□ Absorbing Chains

Definition:

- A state i is **absorbing** if $p_{ii} = 1$.
- A chain is **absorbing** if for every state i there is an absorbing state which is accessible from i .
- A non-absorbing state in an absorbing chain is called a **transient** state.



Decomposition of states

Consider an absorbing chain $\{X_n\}_{n \geq 0}$ with r absorbing states and t transient states.

Re-order the states so that the transient states come first, then the absorbing states. The transition matrix then has the form

$$P = \begin{bmatrix} Q & R \\ 0 & I_r \end{bmatrix}$$

Moreover

$$P^2 = \begin{bmatrix} Q & R \\ 0 & I_r \end{bmatrix} \begin{bmatrix} Q & R \\ 0 & I_r \end{bmatrix} =$$

$$P^3 = P^2 \begin{bmatrix} Q & R \\ 0 & I_r \end{bmatrix} =$$

In general,

$$P^n = \begin{bmatrix} Q^n & M_n \\ 0 & I_r \end{bmatrix}$$

where, $M_n = (I + Q + Q^2 + \cdots + Q^{n-1})R$

Here, for transient states i and j ,

$$(Q^n)_{ij} = \mathbb{P}(X_n = j | X_0 = i)$$

Since j is transient, we get

$$\lim_{n \rightarrow \infty} (Q^n)_{ij} = 0$$

$\lim_{n \rightarrow \infty} Q^n = 0$ implies that $I - Q$ is invertible and

$$\lim_{n \rightarrow \infty} (I + Q + Q^2 + \cdots + Q^{n-1}) = (I - Q)^{-1}$$

So,

$$\lim_{n \rightarrow \infty} P^n = \begin{bmatrix} 0 & (I - Q)^{-1}R \\ 0 & I_r \end{bmatrix} = W$$

- **Expected Number of Visits to Transient States**

Consider an absorbing Markov chain with t transient states.

Theorem: Let i, j be transient states.

- The **Expected Number of Visits** from state i to state j , is given by

$$m_{ij} := ((I - Q)^{-1})_{ij}$$

- The **expected number of steps** of the chain, starting in state i , until it is absorbed, is

$$a_i := \sum_j m_{ij}$$

- The **probability** that the chain is absorbed in state k , given that it started in state i , is

$$b_{ik} := ((I - Q)^{-1}R)_{ik}$$

The matrix $(I - Q)^{-1}$ is called the **fundamental matrix**.

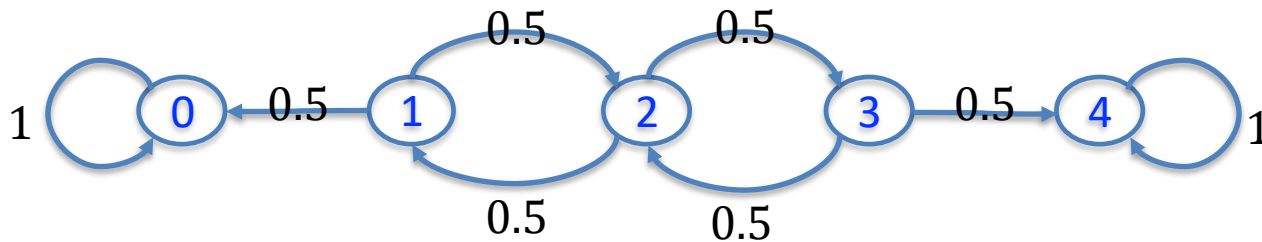
Proof: For a chain starting from transient state $X_0 = i$, the number of visits to transient state j is given by

$$\sum_{k=0}^n \mathbb{I}_{\{X_k=j\}} ,$$

The expected number of visits to j is

$$\begin{aligned} m_{ij} &= E \left[\sum_{k=0}^n \mathbb{I}_{\{X_k=j\}} \right] = \sum_{k=0}^n E[\mathbb{I}_{\{X_k=j\}}] = \sum_{k=0}^n \mathbb{P}(X_k = j \mid X_0 = i) \\ &= \sum_{k=0}^n (P^k)_{ij} = \sum_{k=0}^n (Q^k)_{ij} = ((I - Q)^{-1})_{ij} \end{aligned}$$

Example: For the drunkard's walk,



Under states order $\Omega = \{0, 1, 2, 3, 4\}$,

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0.5 & 0 & 0.5 & 0 & 0 \\ 0 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & 0.5 & 0 & 0.5 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Under decomposition order $\Omega = \{1, 2, 3\} \cup \{0, 4\}$, $P = \begin{bmatrix} Q & R \\ 0 & I_r \end{bmatrix}$

$$Q = \begin{bmatrix} 0 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1/2 & 0 \end{bmatrix}$$

$$R = \begin{bmatrix} 1/2 & 0 \\ 0 & 0 \\ 0 & 1/2 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Compute:

$$Q^2 =$$

$$Q^3 =$$

$$Q^{2n+1} =$$

Example: Gambler's ruin

A gambler starts with \$2 and plays a game in which each round is independent. With probability $p = 0.6$ the gambler wins \$1, and with probability $q = 0.4$ the gambler loses \$1.

The game stops when the gambler either reaches \$5 or is ruined (reaches \$0). Find the probability that the gambler is eventually ruined.

Ordering the states as $(1,2,3,4 \mid 0, 5)$, the transition matrix has the block form

$$P = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix}$$

$$Q = \begin{pmatrix} 0 & 0.6 & 0 & 0 \\ 0.4 & 0 & 0.6 & 0 \\ 0 & 0.4 & 0 & 0.6 \\ 0 & 0 & 0.4 & 0 \end{pmatrix}, \quad R = \begin{pmatrix} 0.4 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0.6 \end{pmatrix}$$

Absorption probabilities is give by the matrix

$$(I - Q)^{-1}R = \begin{pmatrix} 0.616 & 0.384 \\ 0.360 & 0.640 \\ 0.190 & 0.810 \\ 0.076 & 0.924 \end{pmatrix}$$

$$P(\text{ruin} \mid X_0 = 2) = 0.36.$$

Expected Number of Visits is given by matrix

$$(I - Q)^{-1} = \frac{1}{211} \begin{pmatrix} 325 & 285 & 225 & 135 \\ 190 & 475 & 375 & 225 \\ 100 & 250 & 475 & 285 \\ 40 & 100 & 190 & 325 \end{pmatrix}$$

Expected duration until absorption starting from \$2

$$a_2 := \sum_j m_{2j} = \frac{1265}{211} \approx 5.995 \quad \text{steps}$$

Obstruction to Convergence

Each finite irreducible Markov chain has a unique stationary distribution π_*

$$\pi_* P = \pi_*$$

$$\frac{1}{n+1} (I + P + P^2 + \dots + P^n) \xrightarrow{\text{red}} \mathbf{1} \vec{w}^T$$

However, P^n may fail to converge as $n \rightarrow \infty$.

$$P(X_n = k \mid X_0 = i) \rightarrow \pi_k$$

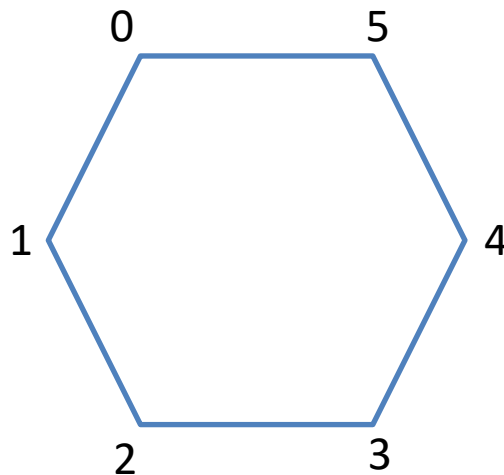
The obstruction is periodicity.

Example: Simple random walk

Take the **simple random walk on the $k = 6$ -cycle C_6** with states

$$\{0,1,2,3,4,5\}(\bmod 6),$$

and from each state you move to the two neighbors with probability $1/2$.



$$\mathbb{P}(X_{t+1} = i + 1 \pmod{6} \mid X_t = i) = \frac{1}{2}$$

$$\mathbb{P}(X_{t+1} = i - 1 \pmod{6} \mid X_t = i) = \frac{1}{2}.$$

Transition matrix P

$$P = \begin{pmatrix} 0 & \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 \end{pmatrix}$$

- **Irreducible**: you can reach any vertex from any other by walking along the cycle.
- **Stationary** distribution: $\pi_* = \left(\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}\right)$
- **Periodic** with period 2: Starting at state i you can return i only at even times.

$$P_{(0,0)}^{2n+1} = \mathbb{P}(X_n = 0 | X_0 = 0) = 0$$

Simple random walk on the $k = 2$ -cycle with transition matrix $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Lab: Chutes/snakes and Ladders

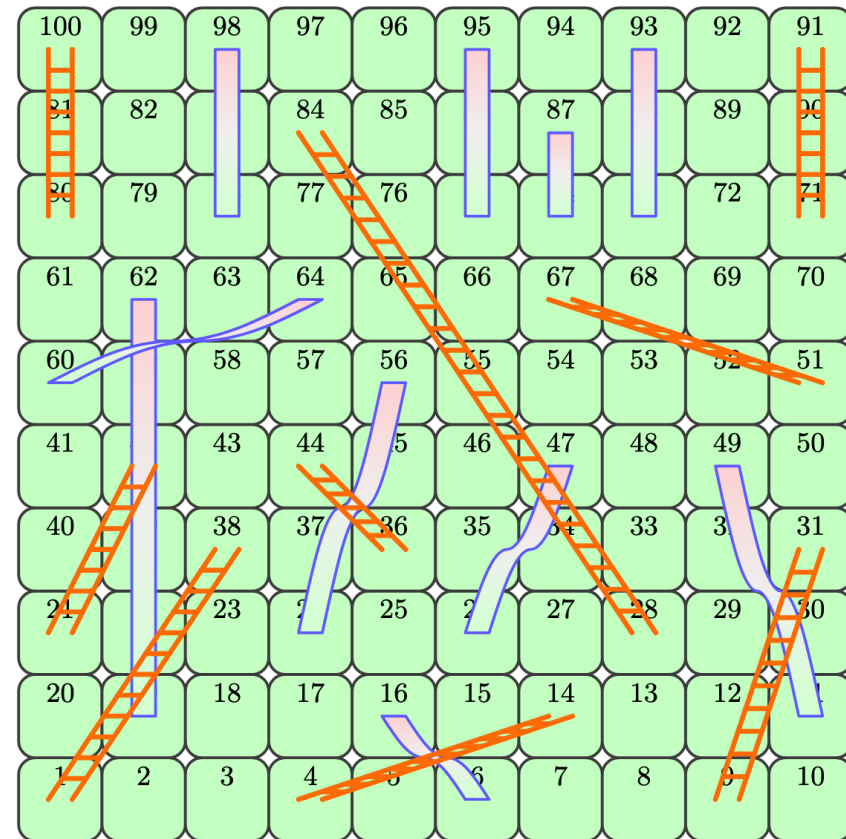
Chutes and Ladders can be modeled as a Markov chain where each state represents the player's current square (from 0 to 100).

On each turn, the player rolls a die and moves forward; landing on a ladder moves them up, while landing on a chute (snake) moves them down.

The next position depends only on the current position and the die roll, so the process is Markovian.

Square 100 is an absorbing state

Find the expected number of turns needed to reach square 100.



□ Aperiodicity.

A chain i is periodic if it can return to a state in a multiple of $d > 1$ steps.

More precisely, consider the set of possible return times to i :

$$\{n > 0 \mid (P^n)_{ii} > 0\}$$

Definition. The period of state i is the greatest common divisor of the set of possible return times to i :

$$d(i) := \gcd\{n > 0 \mid (P^n)_{ii} > 0\}$$

- If $d(i) = 1$, state i is said to be **aperiodic**.
- If $d(i) > 1$, state i is said to be **periodic**.

Simple random walk on the $k = 6$ -cycle

$$d(i) := \gcd\{n > 0 \mid (P^n)_{ii} > 0\} = \gcd\{2, 4, 6, \dots\} = 2$$

$$P^2 = \begin{pmatrix} 0.5 & 0 & 0.25 & 0 & 0.25 & 0 \\ 0 & 0.5 & 0 & 0.25 & 0 & 0.25 \\ 0.25 & 0 & 0.5 & 0 & 0.25 & 0 \\ 0 & 0.25 & 0 & 0.5 & 0 & 0.25 \\ 0.25 & 0 & 0.25 & 0 & 0.5 & 0 \\ 0 & 0.25 & 0 & 0.25 & 0 & 0.5 \end{pmatrix} \quad P^3 = \begin{pmatrix} 0 & 0.375 & 0 & 0.25 & 0 & 0.375 \\ 0.375 & 0 & 0.375 & 0 & 0.25 & 0 \\ 0 & 0.375 & 0 & 0.375 & 0 & 0.25 \\ 0.25 & 0 & 0.375 & 0 & 0.375 & 0 \\ 0 & 0.25 & 0 & 0.375 & 0 & 0.375 \\ 0.375 & 0 & 0.25 & 0 & 0.375 & 0 \end{pmatrix}$$

Although $\mu_n := P(X_n = k \mid X_0 = i)$ does **not** converge, the two subsequences do:

$$\mu_{2k} \longrightarrow \left(\frac{1}{3}, 0, \frac{1}{3}, 0, \frac{1}{3}, 0\right) \quad \mu_{2k+1} \longrightarrow \left(0, \frac{1}{3}, 0, \frac{1}{3}, 0, \frac{1}{3}\right)$$

Cesàro average converges $\frac{1}{N} \sum_{n=0}^{N-1} \mu_n \longrightarrow \left(\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}\right) = \pi.$

Periodicity Is a Class Property

Lemma: All states in a communication class have the same period.

Hence, all states in an irreducible Markov chain have the same period.

Definition:

- A Markov chain is called **periodic**, if it is irreducible and $d > 1$
- A Markov chain is called **aperiodic**, if it is irreducible and $d = 1$

□ Ergodic Markov Chain

Definition: A Markov chain is **ergodic** if it is

- irreducible,
- aperiodic,
- positive-recurrent.

Remark: Finite chains always have finite expected return times (positive-recurrent). So, an irreducible, aperiodic finite-state Markov chain is ergodic.

The transition matrix is regular if and only if the finite chain is ergodic.

Ergodic Theorem

Let $(X_n)_{n \geq 0}$ be a Markov chain on a countable state space Ω with transition matrix P . Assume the chain is

$$\lim_{n \rightarrow \infty} P(X_n = j | X_0 = i) = \pi_j = \frac{1}{r_j} \quad \text{for all } j \in \Omega$$

r_i : Expected number of steps to **return to state** i , starting from i .

The Ergodic Theorem for Markov Chains states that, under certain conditions, the long-run proportion of time a Markov chain spends in a given state converges to the stationary distribution, regardless of the initial state.

Limit Theorem.

Theorem. If $X_1, X_2, \dots$ are realizations from an ergodic Markov chain with stationary distribution π , then:

For any bounded function $h: S \rightarrow \mathbb{R}$,

$$\frac{1}{n} \sum_{i=1}^n h(X_i) \xrightarrow{\text{a.s.}} E_{\pi}\{h(X)\} = \sum_{j=0}^{\infty} h(j)\pi_j \quad \text{as } n \rightarrow \infty$$

In particular, taking $h = \mathbf{1}_{\{j\}}$,

$$\frac{1}{n} \sum_{i=1}^n \mathbf{1}_{(X_i=j)} \xrightarrow{\text{a.s.}} \pi_j \quad \text{as } n \rightarrow \infty$$

So the long-run fraction of time spent in state j equals π_j .

Example: Markov chains have unique limit distribution:

$$T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0.1 & 0.9 \\ 0.6 & 0.4 & 0 \end{bmatrix}$$

If the probability vector for the initial state is $z^{(1)} = (0.5, 0.2, 0.3)$

It follows that $z^{(1)} T = (0.2, 0.6, 0.2)$ and, after several iterations (multiplications by T)

$$z^{(1)} T \rightarrow p(z) = (0.22, 0.41, 0.27)$$

no matter what initial distribution $z^{(1)}$ was.

If this limiting distribution exists, it has to be the stationary distribution.

Example: Symmetric Random Walk on $\mathbb{Z}$:

Markov chain on the state space $\mathbb{Z}$ with the following transition probabilities

$$P(i, i + 1) = P(i, i - 1) = \frac{1}{2} \text{ for any } i \in \mathbb{Z}$$

- The chain is null recurrent.
- There is no stationary distribution.
- The reason to add positive-recurrent is to make sure the unique existence of the stationary distribution

Different definitions of *ergodic* in literature: See [Wikipedia](https://en.wikipedia.org/wiki/Markov_chain#Ergodicity):

https://en.wikipedia.org/wiki/Markov_chain#Ergodicity

Mixing time for ergodic chain

The ergodic Theorem says that

$$\mathbb{P}(X_n = j) \rightarrow \pi_j \text{ as } n \rightarrow \infty \text{ for every state } j.$$

The chain ‘forgets’ about its initial state and settles into the stationary distribution.

How fast does this convergence happen?

Motivating Example: Card Shuffling

[Persi Diaconis](#) famously analyzed the **riffle shuffle** as a Markov chain on the set of all permutations of a 52-card deck.

- **State space:** all 52! permutations
- **Stationary distribution:** uniform over all permutations
- **Initial state:** an ordered deck

After about **7 riffle shuffles**, the deck is already very close to uniform.

Theorem (Rate of Convergence)

Let P be the transition matrix of a finite ergodic Markov chain.

Then there exists a constant $C < \infty$ such that for all $n \geq 1$,

$$\| P^n - \mathbf{1} \vec{w}^T \| \leq C n^{m_2-1} |\lambda_2|^n.$$

Here λ_2 is the second largest eigenvalue of P with algebraic multiplicity m_2 .

□ Time reversible Markov chains

Consider an ergodic chain $\{\dots, X_{n-1}, X_n, X_{n+1}, \dots\}$ with transition probabilities p_{ij} and stationary distribution π . We have

$$p_{ij} = \mathbb{P}(X_n = j \mid X_{n-1} = i) = \mathbb{P}(X_1 = j \mid X_0 = i)$$

Now consider the reversed chain, where we run the sequence backwards:

$$\{\dots, X_{n+1}, X_n, X_{n-1}, \dots\}$$

By Bayes' Theorem, the transition matrix is

$$\begin{aligned} q_{ij} &= P(X_{n-1} = j \mid X_n = i) = \frac{P(X_{n-1} = j, X_n = i)}{P(X_n = i)} = \frac{P(X_n = i \mid X_{n-1} = j)P(X_{n-1} = j)}{P(X_n = i)} \\ &= p_{ji} \frac{P(X_{n-1} = j)}{P(X_n = i)} \end{aligned}$$

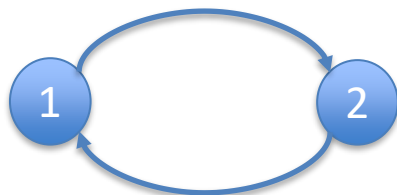
Assume that the original chain is in its stationary distribution $P(X_n = i) = \pi_i$

$$q_{ij} = p_{ji} \frac{\pi_j}{\pi_i}$$

Definition: The Markov chain is **reversible** if

$$q_{ij} = p_{ij} \text{ for all } i, j \in S$$

Example:

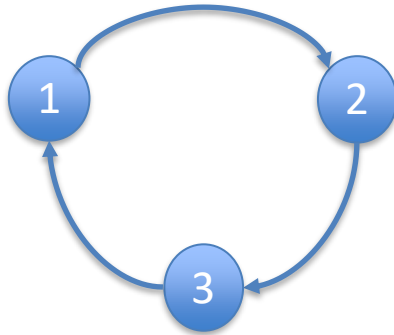


$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Forward: $1 \rightarrow 2 \rightarrow 1 \rightarrow \dots$

Backward: $1 \rightarrow 2 \rightarrow 1 \rightarrow \dots$

Example:



$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

Forward: $1 \rightarrow 2 \rightarrow 3 \rightarrow \dots$

Backward: $1 \rightarrow 3 \rightarrow 2 \rightarrow \dots$

Example:

1

2

3

$$P = \begin{bmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{bmatrix}$$

Is this chain reversible?

Forward: $1 \rightarrow 2 \rightarrow 3 \rightarrow \dots$

Backward: $1 \rightarrow 3 \rightarrow 2 \rightarrow \dots$

The reversible Markov chain condition is equivalent to the “balance equations”:

$$\pi_i p_{ij} = \pi_j p_{ji} \text{ for all } i, j \in S$$

Lemma: Consider an irreducible Markov chain with transition probabilities p_{ij} . Suppose there is a distribution $w_j > 0$ such that for all $i, j \in S$

$$w_i p_{ij} = w_j p_{ji} \text{ for all } i, j \in S$$

Then the chain is time reversible and w_j is the stationary distribution π_j .

So, this result says that if you can find a positive solution of the *simpler* equation then you have solved for the stationary distribution.

Find stationary distribution by solving balance equations

Example:

$$P = \begin{bmatrix} 0 & 1 \\ 2 & 1 \\ \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

Example:



$$P = \begin{bmatrix} 1/3 & 2/3 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 3/4 & 0 & 1/4 & 0 \\ 0 & 0 & 1/5 & 0 & 4/5 \\ 0 & 0 & 0 & 1/2 & 1/2 \end{bmatrix}$$

Try to solve reversible equations:

$$w_i p_{ij} = w_j p_{ji} \text{ for all } i, j \in S$$

Necessary conditions: if $p_{ij} = 0$, then $p_{ji} = 0$.

So, only check the cases where $p_{ij} \neq 0$.

Example: Random walk on weighted graph

Let $G = (V, E)$ be an **undirected** graph. Give each edge $\{i, j\}$ a symmetric weight

$$w_{ij} = w_{ji} \geq 0, \quad w_{ij} > 0 \text{ iff } \{i, j\} \in E.$$

Define the **weighted degree**

$$d_i := \sum_{k \in V} w_{ik}$$

Define the random walk **transition probabilities**

$$p_{ij} := \frac{w_{ij}}{d_i}$$

(So from i , you move to a neighbor j with probability proportional to the edge weight.)

Assume the graph is connected so the chain is irreducible.

Solve detailed balance

$$\pi_i p_{ij} = \pi_j p_{ji}$$

$$\pi_i \frac{w_{ij}}{d_i} = \pi_j \frac{w_{ji}}{d_j}$$

Here $w_{ij} = w_{ji}$

$$\frac{\pi_i}{d_i} = \frac{\pi_j}{d_j}$$

π_i is proportional to d_i , and we also need $\sum_i \pi_i = 1$

So, we can try

$$\pi_i := \frac{d_i}{total} = \frac{d_i}{\sum_v d_v}$$

So detailed balance holds, hence the chain is **reversible** and π is **stationary**.

Example: Random Walk on a Graph

Consider a random walk on an undirected graph.

Transition

$$p_{ij} = \begin{cases} \frac{1}{\deg(i)} & \text{if } i \sim j \\ 0 & \text{otherwise.} \end{cases}$$

Stationary distribution

$$\pi_i = \frac{\deg(i)}{\sum_k \deg(k)} = \frac{\deg(i)}{2|E|}$$

The quantity $\pi_i p_{ij}$ has another interpretation: it is the rate of jumps of the chain from state i to state j . More precisely, it is the long-run average rate at which the chain makes the transition between these states:

$$\lim_{n \rightarrow \infty} P(X_n = i, X_{n+1} = j) = \pi_i p_{ij}$$

For the Random Walk on a Graph example

$$\pi_i p_{ij} = \frac{d_i}{\sum_v d_v} \frac{w_{ij}}{d_i} = \frac{w_{ij}}{\sum_v d_v}$$

is the long-run **probability flux** across the edge $i \rightarrow j$.

On an undirected graph, the flux is symmetric: flux $i \rightarrow j$ equals flux $j \rightarrow i$. That symmetry *is* reversibility.

If the graph is **directed** or weights aren't symmetric, detailed balance typically fails, and the stationary distribution usually can't be read off from degrees.

❖ References:

Recommended textbooks

- Introduction to Probability Models - Sheldon M. Ross
- Probability and Random Processes - Geoffrey Grimmett and David Stirzaker
- Introduction to Probability, Statistics, and Random Processes - Hossein Poshro-Nik
- Introduction to Stochastic Processes with R - Robert Dobrow

More textbooks:

- Essentials of Stochastic Processes – Richard Durrett
- Understanding Markov Chains- Nicolas Privault
- Markov Chains –J.R. Norris
- Stochastic Processes, an Introduction. –Peter Jones and Peter Smith
- Stochastic Processes 2nd Edition by Sheldon M. Ross